ProjectInitializationandPlanningPhase

Date	22April2026
MemberID	Omkar Bhimrao Magdum
ProjectTitle	RestaurantRecommendation System
MaximumMarks	3 Marks

ProjectProposal(ProposedSolution)

Thisprojectproposaloutlinesasolutiontoaddressaspecificproblem.Withaclearobjective,defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	To develop a system that provides personalized and efficient restaurant recommendations by analyzing user preferences, dietary requirements, location, and budget.
Scope	The project aims to serve users seeking restaurant suggestions that match their individual lifestyle choices and dining preferences. It will operate across various regions, considering real-time data and qualitative reviews.
Problem Statement	
Description	Finding restaurants tailored to specific needs is often time-consuming and inefficient. Users frequently revisit the same places, missing diverse options that better match their preferences.
Impact	Solving this problem improves user satisfaction, encourages exploration of new dining options, and reduces time spent on decision-making.
Proposed Solution	
Approach	The solution employs innovative recommendation algorithms that factor in both user input and external data like ambiance, ratings, and reviews. It adapts dynamically to user feedback and real-time changes.
Key Features	• Personalized recommendations • Real-time data analysis • Integration of user reviews • Consideration of dietary and budget constraints • Scalable infrastructure

ResourceRequirements

ResourceType	Description	Specification/Allocation
Hardware		
ComputingResources	8-coreCPUsandoptionalGPU	2xNVIDIAV100GPUs
Memory	RAM	Minimum8GB RAM
Storage	SSD	1TBSSDforstoringuserdata and restaurant metadata
Software		
Frameworks	Python frameworks	Python, Flask
Libraries	Additionallibraries	Pandas,NumPy,Scikit-learn, TensorFlow, BeautifulSoup (for scraping),andNLTK(forreview analysis)
DevelopmentEnvironment	IDE,versioncontrol	Jupyter Notebook
Data		
Data	Size:-Approx.50,000–100,000 recordsinitially;scalablebased on user growth,	Aggregatedfromcrowdsourced restaurantplatforms(e.g.,Yelp, Zomato APIs), user feedback, and public review datasets
	Format: - CSV for tabular datasets,Text/HTMLforscraped reviews	